Hospital Billing and Patient Classification System

Problem Statement: Process consultation, diagnostic, and treatment charges, apply conditional concessions, validate billing constraints, calculate the final bill, and classify patients according to the payment policy.

1. Computational Thinking

1. Decomposition: Divide the problem into smaller tasks:

- Get patient details.
- Calculate consultation charges.
- Calculate diagnostic charges.
- Calculate treatment charges.
- Apply concessions.
- Calculate final bill.
- Classify patient based on payment.

2. Pattern Recognition:

- Every patient has a unique patient ID.
- All patients have different medical charges.
- Discounts depend on eligibility conditions.
- Payment status determines patient classification.

3. Abstraction:

- Consider only patient details, charges, concessions, and payment status.
- Ignore unnecessary medical information.

4. Algorithm Design:

- Input patient details and charges.
- Validate the entered values.
- Calculate total charges.
- Apply eligible concessions.
- Calculate final amount.
- Check payment status.
- Display the bill and classification.

2. Algorithm

1. Start.
2. Read patient name and patient ID.
3. Read consultation, diagnostic, and treatment charges.
4. Validate that all charges are non-negative.
5. Calculate total bill.

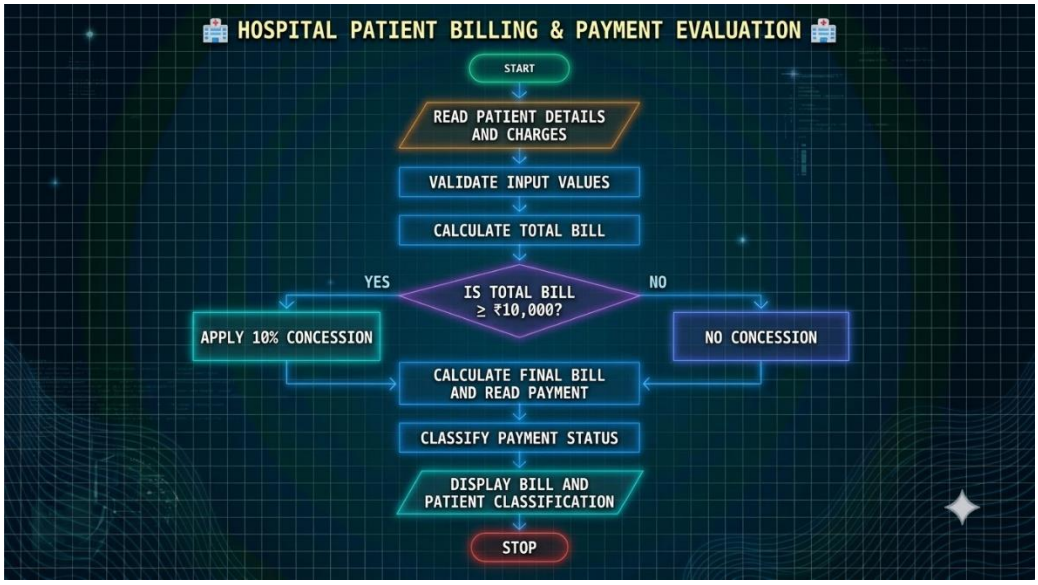
Total = Consultation + Diagnostic + Treatment

6. Check concession eligibility.
7. If eligible, calculate the discount; otherwise, discount is zero.
8. Calculate final bill.

Final Bill = Total - Discount

9. Read amount paid.
10. Validate that payment is between zero and the final bill.
11. Classify the patient:
    - If paid amount = final bill → Fully Paid.
    - If paid amount > 0 → Partially Paid.
    - If paid amount = 0 → Unpaid.
12. Display patient details, total bill, discount, final bill, payment status, and balance.
13. Stop.

3. Flowchart:



4. Variables

Variable	Purpose
patient_id	Patient identification
patient_name	Patient name
consultation	Consultation charge
diagnostic	Diagnostic charge
treatment	Treatment charge
total	Total charges
discount	Concession amount
final_bill	Final amount
paid	Amount paid
balance	Remaining amount
status	Payment classification

5. Data Types

Data type	Variables
Integer	patient_id

String	patient_name, status
Float	consultation, diagnostic, treatment, total, discount, final_bill, paid, balance
Boolean	concession_eligible

6. Operators

- Arithmetic: +, -, *, /
- Relational: >, <, ==, >=
- Logical: and, or, not
- Assignment: =, +=

7. Conditional Statements

- if: Check concession eligibility.
- elif: Check payment classification.
- else: Handle other conditions.

Example:

if paid == final_bill: status = "Fully Paid"elif paid > 0: status = "Partially Paid"else: status = "Unpaid"

8. Loop

Use a for loop to process multiple patients.

for i in range(n): print("Enter patient details")

9. Strings

Strings are used to store patient names and payment status.

Example:

patient_name = "Ravi"status = "Fully Paid"print(patient_name, status)

10. Pattern Generation

Generate a simple bill receipt pattern.

for i in range(1, 6): print("*" * i)

Output:

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11. Lists

Lists store multiple patients and their billing amounts.

patients = ["Ravi", "Sita", "Kiran"]bills = [5000, 3000, 4500]print(patients)print(bills)

12. Tuples

Tuples store fixed billing information.

charges = (500, 1000, 2500)print("Consultation:", charges[0])print("Diagnostic:", charges[1])print("Treatment:", charges[2])

13. Packing and Unpacking

Packing: Combining multiple values into a tuple.

bill = (500, 1000, 2500)

Unpacking: Assigning tuple values to separate variables.

consultation, diagnostic, treatment = billprint(consultation, diagnostic, treatment)

14. Pseudocode

START

INPUT number_of_patients

FOR each patient

  INPUT patient_id

  INPUT patient_name

  INPUT consultation

  INPUT diagnostic

  INPUT treatment

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IF any charge < 0 THEN
   PRINT "Invalid charges"
   CONTINUE
END IF

total = consultation + diagnostic + treatment

INPUT concession_eligibility

IF concession_eligibility == "yes" THEN
   discount = total * 0.10
ELSE
   discount = 0
END IF

final_bill = total - discount

INPUT paid

IF paid < 0 OR paid > final_bill THEN
   PRINT "Invalid payment"
   CONTINUE
END IF

balance = final_bill - paid

IF paid == final_bill THEN
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        status = "Fully Paid"
    ELSE IF paid > 0 THEN
        status = "Partially Paid"
    ELSE
        status = "Unpaid"
    END IF

    PRINT patient_name
    PRINT total
    PRINT discount
    PRINT final_bill
    PRINT balance
    PRINT status

END FOR

STOP
```

15. Sample Output

HOSPITAL BILL RECEIPT

Patient Name: Ravi

Patient ID: 101

Description	**Amount**
Consultation	₹500
Diagnostic	₹1,000
Treatment	₹2,500
Total	₹4,000
Discount (10%)	₹400

Final Bill	₹3,600
Amount Paid	₹2,000
Balance	₹1,600

Payment Status

Partially Paid

16. Conclusion

The Hospital Billing System uses computational thinking, algorithms, flowcharts, variables, data types, operators, conditional statements, loops, strings, pattern generation, lists, tuples, and packing/unpacking to calculate hospital bills accurately and classify patients according to their payment status.